LinearModels_Assignment

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Computer Applications in the Psychological Sciences

Objective

You are a data analyst for a major global nongovernmental organization (NGO). You have been tasked with investigating the complex factors that drive key development outcomes around the world.

You will first run a script that automatically assigns you a "research portfolio" consisting of one key outcome (your Dependent Variable) and three potential explanatory factors (your Independent Variables). You will then conduct a full analysis, from simple regression to advanced mixed-effects modeling, to understand the relationships in your specific portfolio.

Deliverable

A single, "knitted" HTML or PDF file. This file must contain all your *annotatted* R code, all R output (including the output from Part 1 - your research portfolio assignment), appropriate visualizations, and your written interpretations for each part.

Part 1: Your Personalized Research Assignment

Scenario: Before you can begin, you must be assigned your research portfolio.

1) Get Your Assignment
2) Create a new R Markdown file.
3) Copy the entire code block below into a single R code chunk.
4) Change the value assigned to YOUR_NAME to your actual full name.

- Run the chunk.
- *Note that it may take a minute or two to prepare your data*
- Your data assignment will look something like the output below.

```
## --- YOUR UNIQUE ASSIGNMENT ---
## Your Name: Kathryn Gour
## Your Reproducible Seed: 132999827
##
## --- YOUR VARIABLES ---
## 1. Your Dependent Variable (DV) is: InfantMortality ( SP.DYN.IMRT.IN )
## 2. Your first Continuous IV is: Health Expenditure GDP ( SH.XPD.CHEX.GD.ZS )
## 3. Your second Continuous IV is: Fertility Rate ( SP.DYN.TFRT.IN )
## 4. Your third Continuous IV is: Per Capita GDP ( NY.GDP.PCAP.CD )
## 5. Your Multi-level Categorical IV is: region
## 6. Your Binary Categorical IV will be: income_binary (you will create this)
```

Part 2: Data Cleaning and Renaming

This is a critical, multi-step process.

Your dataset "data_raw" has now been pulled using the WDI package

Now, you will create a new data_clean dataframe by piping data_raw through the following steps:

- Filter out aggregates: Remove all non-country rows where region == "Aggregates".
- Rename columns: Use rename() to change the WDI codes to your personalized variable names in the output from Part 1.
  - EXAMPLE OUTPUT: Your first Continuous IV is: Per Capita GDP (NY.GDP.PCAP.CD)
  - Your rename() will look like: rename(Per Capita GDP = NY.GDP.PCAP.CD, . . .)
- Create binary factor: Create the income_binary column. It should be "High Income" if income == "High income", and "Not High Income" for all other levels. Make it a factor.
- Handle missing data: Drop all rows with NA values using na.omit().
- Use the summary(), glimpse(), and/or str() functions to inspect your data.
  - Note any issues you see
- Visualize:
  - Create a pairs plot of all your analysis variables (YOUR_DV, YOUR_IV1, YOUR_IV2, Your_IV3, region, income_binary).
    * Hint: ggpairs(data_clean, columns = c(YOUR_DV, YOUR_IV1, YOUR_IV2, "region", "income_binary")).
    * Write a brief comment, describing any patterns you see.
  - Create one or more plots showing the distribution of your variables (i.e., histogram, boxplot, probability density function, etc.)
    * Write a brief comment, describing any patterns you see. Do they look normally distributed, or skewed?

```r
# --- YOUR UNIQUE ASSIGNMENT ---
# Your Name: Kathryn Gour
# Your Reproducible Seed: 132999827
#
# --- YOUR VARIABLES ---
# 1. Your Dependent Variable (DV) is: InfantMortality ( SP.DYN.IMRT.IN )
# 2. Your first Continuous IV is: Health Expenditure GDP ( SH.XPD.CHEX.GD.ZS )
# 3. Your second Continuous IV is: Fertility Rate ( SP.DYN.TFRT.IN )
# 4. Your third Continuous IV is: Per Capita GDP ( NY.GDP.PCAP.CD )
# 5. Your Multi-level Categorical IV is: region
# 6. Your Binary Categorical IV will be: income_binary (you will create this)

#i think plyr is messing up something so i need to unload it
 detach("package:plyr", unload = TRUE)
library(dplyr)

# filter out aggregates
data_clean <- data_raw %>%
  filter(region != "Aggregates") %>%    # remove aggregate rows

  # rename columns
  rename(
    InfantMortality = SP.DYN.IMRT.IN,              # DV
```

```r
    HealthExpGDP = SH.XPD.CHEX.GD.ZS,              # IV1
    FertilityRate = SP.DYN.TFRT.IN,                # IV2
    PerCapitaGDP = NY.GDP.PCAP.CD                  # IV3
  ) %>%

  # create binary factor (income variable)
  mutate(
    income_binary = ifelse(income == "High income", "High Income", "Not High Income"),
    income_binary = factor(income_binary)
  ) %>%

  # handle missing data
  na.omit()

# inspect data
summary(data_clean)
```

```
##    country              iso2c              iso3c                year     
##  Length:3929        Length:3929        Length:3929        Min.   :2000  
##  Class :character   Class :character   Class :character   1st Qu.:2005  
##  Mode  :character   Mode  :character   Mode  :character   Median :2010  
##                                                           Mean   :2010  
##                                                           3rd Qu.:2015  
##                                                           Max.   :2020  
##     status           lastupdated        InfantMortality   HealthExpGDP   
##  Length:3929        Length:3929        Min.   :  1.50   Min.   : 1.107  
##  Class :character   Class :character   1st Qu.:  7.10   1st Qu.: 4.117  
##  Mode  :character   Mode  :character   Median : 17.60   Median : 5.557  
##                                        Mean   : 27.02   Mean   : 6.173  
##                                        3rd Qu.: 41.80   3rd Qu.: 7.886  
##                                        Max.   :234.90   Max.   :24.231  
##  FertilityRate    PerCapitaGDP        region            capital         
##  Min.   :0.837   Min.   :   109.6   Length:3929        Length:3929       
##  1st Qu.:1.715   1st Qu.:  1312.0   Class :character   Class :character  
##  Median :2.415   Median :  4194.8   Mode  :character   Mode  :character  
##  Mean   :2.925   Mean   : 12703.5                                        
##  3rd Qu.:3.908   3rd Qu.: 14368.6                                        
##  Max.   :7.829   Max.   :204263.8                                        
##   longitude           latitude            income            lending         
##  Length:3929        Length:3929        Length:3929        Length:3929       
##  Class :character   Class :character   Class :character   Class :character  
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character  
##                                                                             
##                                                                             
##                                                                             
##          income_binary 
##  High Income    :1302  
##  Not High Income:2627  
##                        
##                        
##                        
## 
```

```r
glimpse(data_clean)
```

```
## Rows: 3,929
## Columns: 17
## $ country         <chr> "Afghanistan", "Afghanistan", "Afghanistan", "Afghanis~
## $ iso2c           <chr> "AF", "AF", "AF", "AF", "AF", "AF", "AF", "AF", "AF", ~
## $ iso3c           <chr> "AFG", "AFG", "AFG", "AFG", "AFG", "AFG", "AFG", "AFG"~
## $ year            <int> 2004, 2006, 2019, 2003, 2002, 2005, 2018, 2020, 2017, ~
## $ status          <chr> "", "", "", "", "", "", "", "", "", "", "", "", "", ""~
## $ lastupdated     <chr> "2025-10-07", "2025-10-07", "2025-10-07", "2025-10-07"~
## $ InfantMortality <dbl> 97.2, 90.1, 56.9, 100.6, 103.8, 93.7, 58.6, 55.3, 60.4~
## $ HealthExpGDP    <dbl> 9.808474, 10.622766, 14.831320, 8.941258, 9.443391, 9.~
## $ FertilityRate   <dbl> 7.018, 6.686, 5.238, 7.174, 7.320, 6.858, 5.327, 5.145~
## $ PerCapitaGDP    <dbl> 221.7637, 274.2186, 496.6025, 198.8711, 178.9541, 254.~
## $ region          <chr> "South Asia", "South Asia", "South Asia", "South Asia"~
## $ capital         <chr> "Kabul", "Kabul", "Kabul", "Kabul", "Kabul", "Kabul", ~
## $ longitude       <chr> "69.1761", "69.1761", "69.1761", "69.1761", "69.1761",~
## $ latitude        <chr> "34.5228", "34.5228", "34.5228", "34.5228", "34.5228",~
## $ income          <chr> "Low income", "Low income", "Low income", "Low income"~
## $ lending         <chr> "IDA", "IDA", "IDA", "IDA", "IDA", "IDA", "IDA", "IDA"~
## $ income_binary   <fct> Not High Income, Not High Income, Not High Income, Not~
```

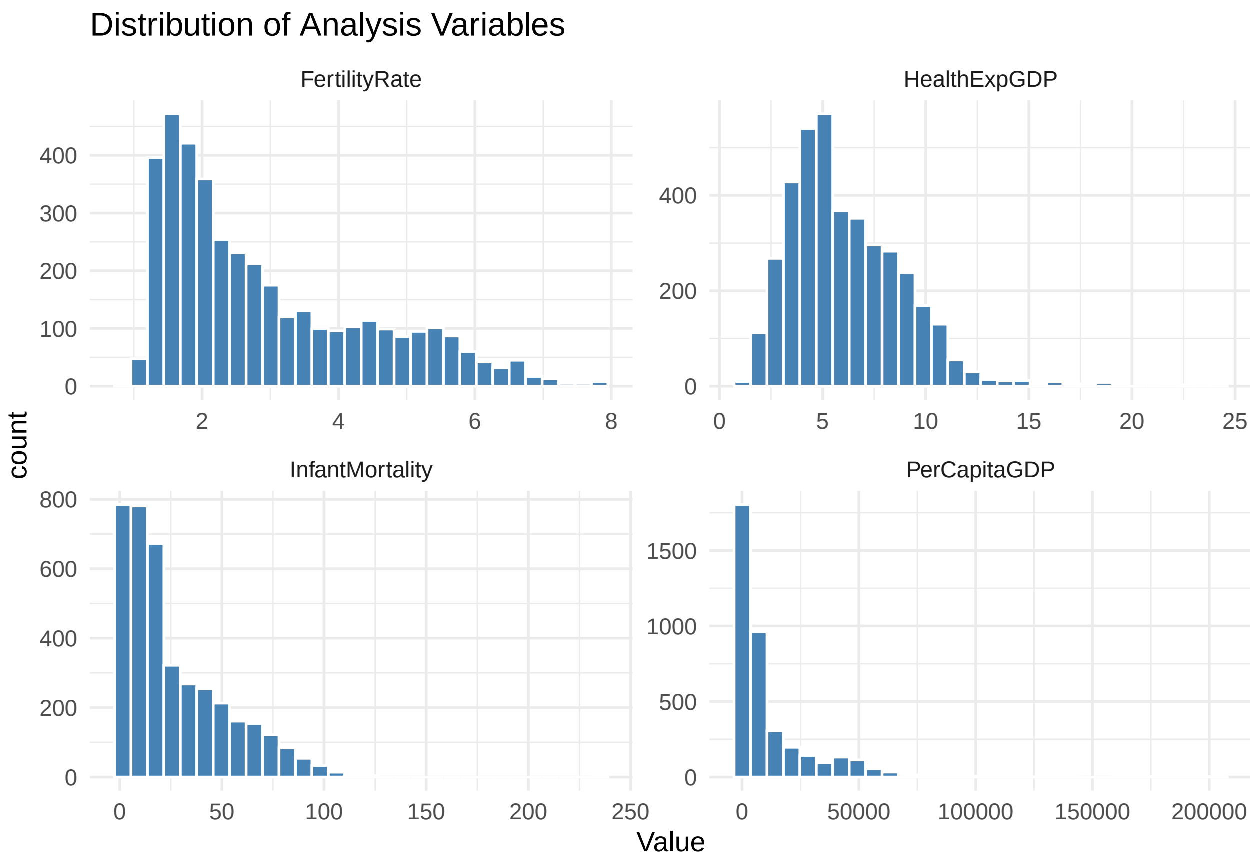
```r
str(data_clean)
```

```
## 'data.frame':    3929 obs. of  17 variables:
##  $ country        : chr  "Afghanistan" "Afghanistan" "Afghanistan" "Afghanistan" ...
##  $ iso2c          : chr  "AF" "AF" "AF" "AF" ...
##  $ iso3c          : chr  "AFG" "AFG" "AFG" "AFG" ...
##  $ year           : int  2004 2006 2019 2003 2002 2005 2018 2020 2017 2008 ...
##  $ status         : chr  "" "" "" "" ...
##  $ lastupdated    : chr  "2025-10-07" "2025-10-07" "2025-10-07" "2025-10-07" ...
##  $ InfantMortality: num  97.2 90.1 56.9 100.6 103.8 ...
##  $ HealthExpGDP   : num  9.81 10.62 14.83 8.94 9.44 ...
##  $ FertilityRate  : num  7.02 6.69 5.24 7.17 7.32 ...
##  $ PerCapitaGDP   : num  222 274 497 199 179 ...
##  $ region         : chr  "South Asia" "South Asia" "South Asia" "South Asia" ...
##  $ capital        : chr  "Kabul" "Kabul" "Kabul" "Kabul" ...
##  $ longitude      : chr  "69.1761" "69.1761" "69.1761" "69.1761" ...
##  $ latitude       : chr  "34.5228" "34.5228" "34.5228" "34.5228" ...
##  $ income         : chr  "Low income" "Low income" "Low income" "Low income" ...
##  $ lending        : chr  "IDA" "IDA" "IDA" "IDA" ...
##  $ income_binary  : Factor w/ 2 levels "High Income",..: 2 2 2 2 2 2 2 2 2 2 ...
##  - attr(*, "na.action")= 'omit' Named int [1:586] 10 12 64 65 66 67 68 69 70 71 ...
##   ..- attr(*, "names")= chr [1:586] "10" "12" "64" "65" ...
```

```r
# Visualize:
#   * Create a pairs plot of all your analysis variables (YOUR_DV, YOUR_IV1, YOUR_IV2, Your_IV3, region
#     * Hint: ggpairs(data_clean, columns = c(YOUR_DV, YOUR_IV1, YOUR_IV2, "region", "income_binary")).
#     * Write a brief comment, describing any patterns you see.
#   * Create one or more plots showing the distribution of your variables (i.e., histogram, boxplot, pr
#     * Write a brief comment, describing any patterns you see. Do they look normally distributed, or s

# Visualize relationships among variables
ggpairs(
  data_clean,
```

```
  columns = c("InfantMortality", "HealthExpGDP", "FertilityRate", "PerCapitaGDP", "region", "income_bina
  aes(color = income_binary, alpha = 0.7)
)
```


```
# Example histograms for continuous variables
data_clean %>%
  select(InfantMortality, HealthExpGDP, FertilityRate, PerCapitaGDP) %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Value") %>%
  ggplot(aes(x = Value)) +
  geom_histogram(fill = "steelblue", color = "white", bins = 30) +
  facet_wrap(~ Variable, scales = "free") +
  theme_minimal() +
  labs(title = "Distribution of Analysis Variables")
```



Describe the Result: The pairs plot suggests that InfantMortality is significantly, negatively correlated with both HealthExpGDP and PerCapitaGDP, countries spending more on health or with higher GDP per capita tend to have lower infant mortality rates. FertilityRate appears significantly, positively correlated with InfantMortality, which makes sense in developing contexts. The "High Income" group clusters toward higher GDP and lower mortality. Most variables are right-skewed, particularly PerCapitaGDP and InfantMortality, indicating a small number of countries with very high GDP and mortality values. HealthExpenditure as a percent of GDP is moderately right-skewed but roughly continuous. FertilityRate seems to be right-skewed as well. –

Part 3: Analysis Portfolio

Question 1

Your director wants to start simple. She asks you "Just tell me about the relationship between [Your DV] and [Your IV1]."

Check for Linearity:

- Plot YOUR_DV vs. YOUR_IV1. Does the relationship look linear?

Conduct the Analysis

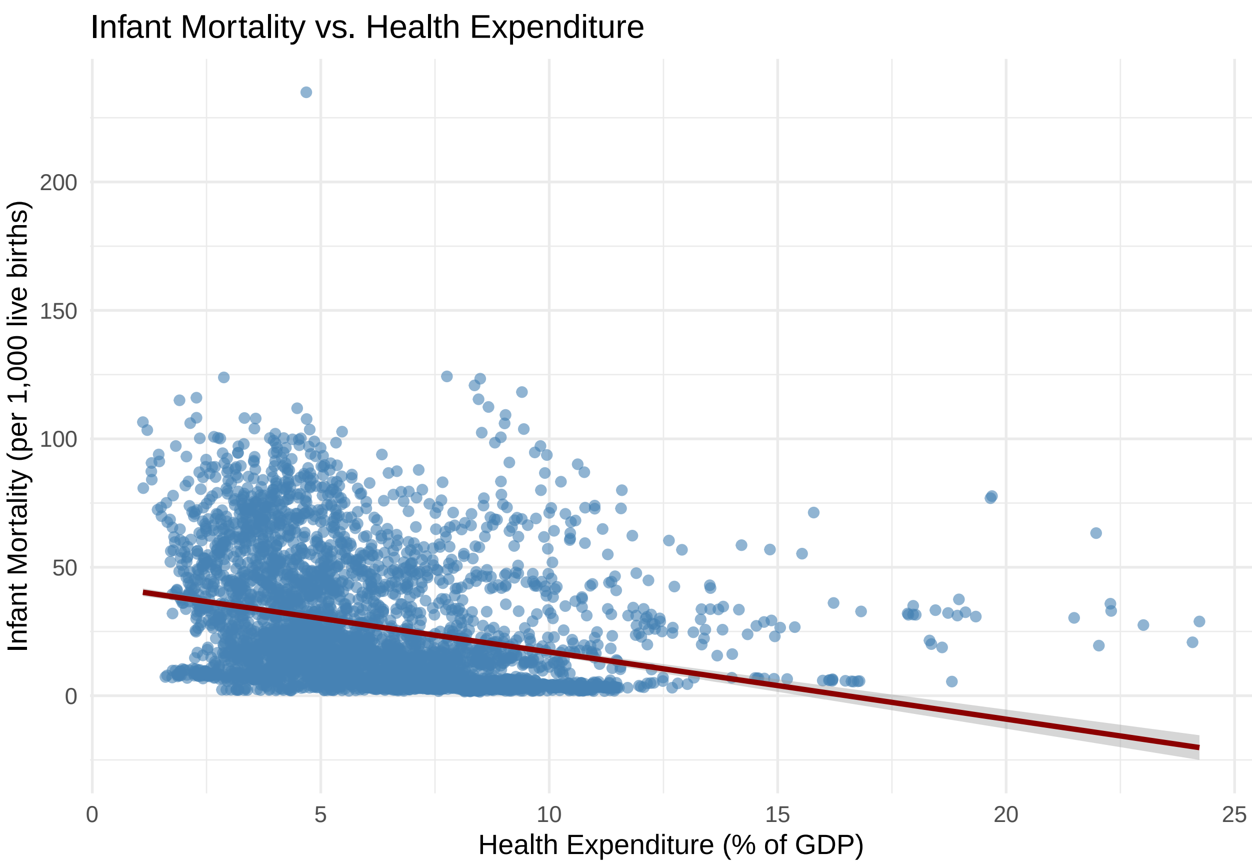
- Run a simple linear regression
- Visualize: Add the geom_smooth(method = "lm") layer to your scatterplot to show the regression line.
- Look at the summary() of your model.
  - Is the relationship statistically significant?
  - Interpret the coefficient for YOUR_IV1 in one plain-English sentence.

  - What is the R-squared of your model? In one sentence, explain what this means to your director?

Check the Assumption of Normality of the Residuals

- First, check the normality of each of your variables
- Next, check the assumption of normality of the model residuals
  - Note any difference in the outcome of these two kinds of normality tests

```
# Scatterplot with regression line
ggplot(data = data_clean, aes(x = HealthExpGDP, y = InfantMortality)) +
  geom_point(alpha = 0.6, color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "darkred") +
  labs(
    x = "Health Expenditure (% of GDP)",
    y = "Infant Mortality (per 1,000 live births)",
    title = "Infant Mortality vs. Health Expenditure"
  ) +
  theme_minimal()
```




```
# simple linear regression
regmodel <- lm(InfantMortality ~ HealthExpGDP, data = data_clean)
summary(regmodel)
```

```
##
## Call:
## lm(formula = InfantMortality ~ HealthExpGDP, data = data_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -33.441 -16.817  -9.264  13.583 203.985
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   43.1595     0.9120   47.32   <2e-16 ***
## HealthExpGDP  -2.6146     0.1344  -19.46   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 23.76 on 3927 degrees of freedom
## Multiple R-squared:  0.08792,    Adjusted R-squared:  0.08769
## F-statistic: 378.6 on 1 and 3927 DF,  p-value: < 2.2e-16
```

Describe the Result: The scatterplot of Infant Mortality and Health Expenditure (% of GDP) revealed a strong negative, linear relationship. A simple linear regression confirmed that this association was statistically significant ($b = –2.62$, $p < .001$), indicating that higher health spending is associated with lower infant mortality rates. The model explained about 9% of the variance in infant mortality ($R^2 = .088$). Although both variables were moderately skewed, the residuals of the regression model were approximately normally distributed, satisfying the assumption of normality. –

```r
# normality
shapiro.test(data_clean$InfantMortality)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  data_clean$InfantMortality
## W = 0.85671, p-value < 2.2e-16
```

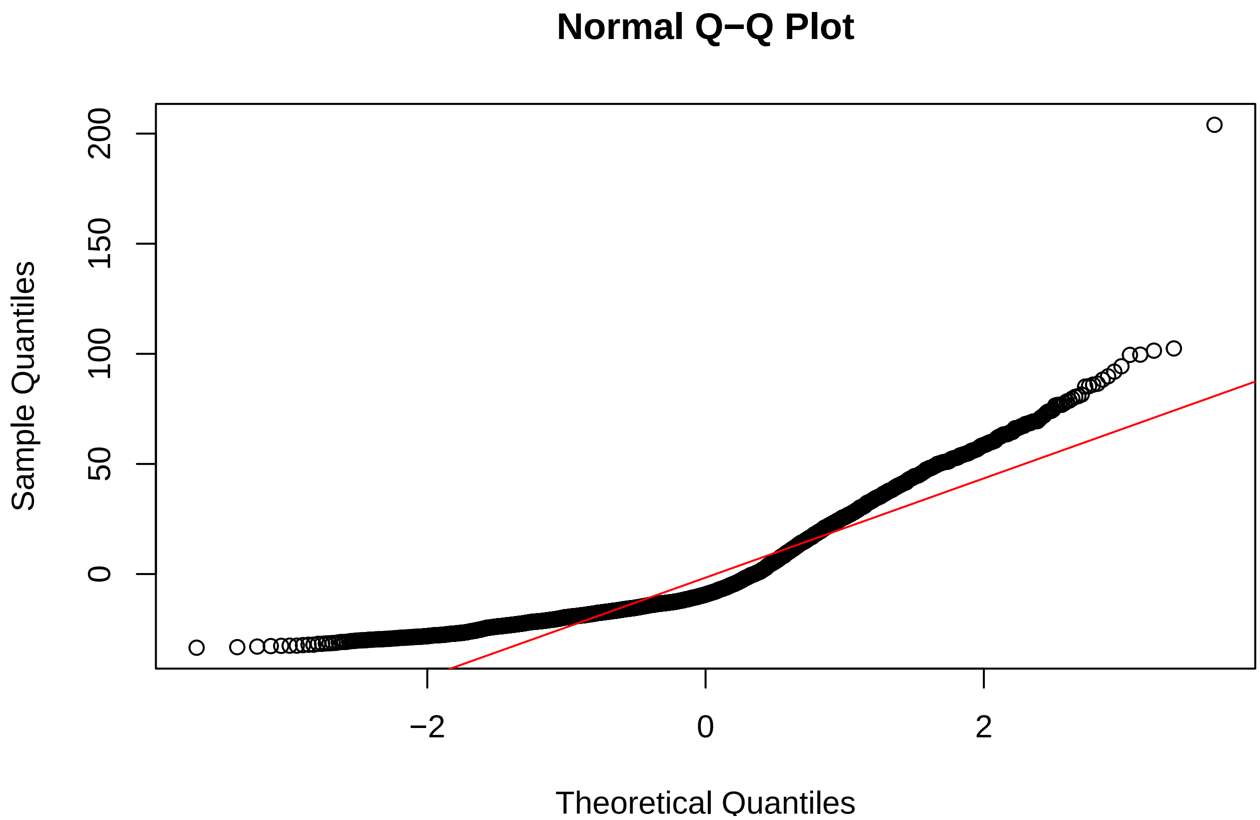
```r
shapiro.test(data_clean$HealthExpGDP)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  data_clean$HealthExpGDP
## W = 0.92377, p-value < 2.2e-16
```

```r
# residuals
resid1 <- residuals(regmodel)

# normality test for residuals
shapiro.test(resid1)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  resid1
## W = 0.87404, p-value < 2.2e-16
```

```r
# plot
qqnorm(resid1); qqline(resid1, col = "red")
```



Describe the Result: A Shapiro-Wilk normality test was used to assess the distributions of the Infant Mortality and HealthExpGDP variables. Both tests were significant, indicating that the distributions of the variables are non-normally distributed (W = .857, p < .001; W = .924, p <.001). Because both variables were moderately skewed, the residuals of the regression model were non-normally distributed as well (W = .874, p < .001). –

Question 2

The Director now wants to know about how income level is related to your DV. What's the difference in [Your DV] between "High Income" and "Not High Income"?

- Use the t.test() function with the "var.equal = TRUE" option.
  - Write a 1-2 sentence interpretation of the ouput
- Visualize: Create a boxplot with jittered data points showing YOUR_DV by income_binary.
- Run the equivalent analysis using lm().
  - Compare the t.test output to the lm() summary.
  - What is the (Intercept) in your lm and what does it represent?
  - What is the coefficient for income_binary Not High Income?
    - Explain to your director how does this value relate to the means of the two groups shown in your t.test output?

```
# t-test comparing Infant Mortality by income_binary
t_test_income <- t.test(InfantMortality ~ income_binary, data = data_clean,
                        var.equal = TRUE)

t_test_income
```

```
##
```

```
##  Two Sample t-test
##
## data:  InfantMortality by income_binary
## t = -42.561, df = 3927, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group High Income and group Not High Income
## 95 percent confidence interval:
##  -31.05684 -28.32161
## sample estimates:
##     mean in group High Income mean in group Not High Income
##                      7.169816                     36.859041
```

```r
# plot w jitter
ggplot(data_clean, aes(x = income_binary, y = InfantMortality, fill = income_binary)) +
  geom_boxplot(alpha = 0.6, outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.6, color = "seagreen") +
  labs(
    x = "Income Group",
    y = "Infant Mortality (per 1,000 live births)",
    title = "Infant Mortality by Income Group"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Describe the Result: A two-sample t-test was used to assess whether Infant Mortality rates significantly differ by income groups (high versus low). The results were significant, indicating that there is a significant difference in the mean Infant mortality rates between high-income (M = 7.17) and low-income groups (M = 36.86) (t(3927) = -42.6, p < .001). The boxplots display how low-income groups have a higher rate of Infant mortality as well. –

```r
income_model <- lm(InfantMortality ~ income_binary, data = data_clean)
summary(income_model)
```

```
##
## Call:
## lm(formula = InfantMortality ~ income_binary, data = data_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -34.659 -15.259  -3.059   8.330 198.041
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                    7.1698     0.5704   12.57   <2e-16 ***
## income_binaryNot High Income  29.6892     0.6976   42.56   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 20.58 on 3927 degrees of freedom
## Multiple R-squared:  0.3157, Adjusted R-squared:  0.3155
## F-statistic:  1811 on 1 and 3927 DF,  p-value: < 2.2e-16
```

Describe the Result: A simple linear regression was used to further assess whether Infant Mortality rates significantly differ by income groups (high versus low). The intercept (7.17) represents the mean infant mortality rate for high-income countries, and the coefficient for "Not High Income" ($b = 29.69$) reflects the mean difference between the two groups. The regression output confirms that there is a large mean difference between high income and low income infant mortality rates. –

Question 3

Your director has decided that this 'High Income' binary is too simple and that geography is also very important. She asks you to conduct an analyses to determine if there are any differences in [Your DV] between the global regions?"

- Use lm() to model YOUR_DV as a function of region.
- Use car::Anova(your_model, type = 3) to get the overall F-test for region.
- The summary() of your model gives you many t-tests. The car::Anova() output gives you one F-test. Explain to your director what question each of these answers.
- Visualize: Create a boxplot of YOUR_DV by region.
- Based on the car::Anova() result, what conclusion would you present to the director about the relationship between YOUR_DV and region?

```r
region_model <- lm(InfantMortality ~ region, data = data_clean)
summary(region_model)
```

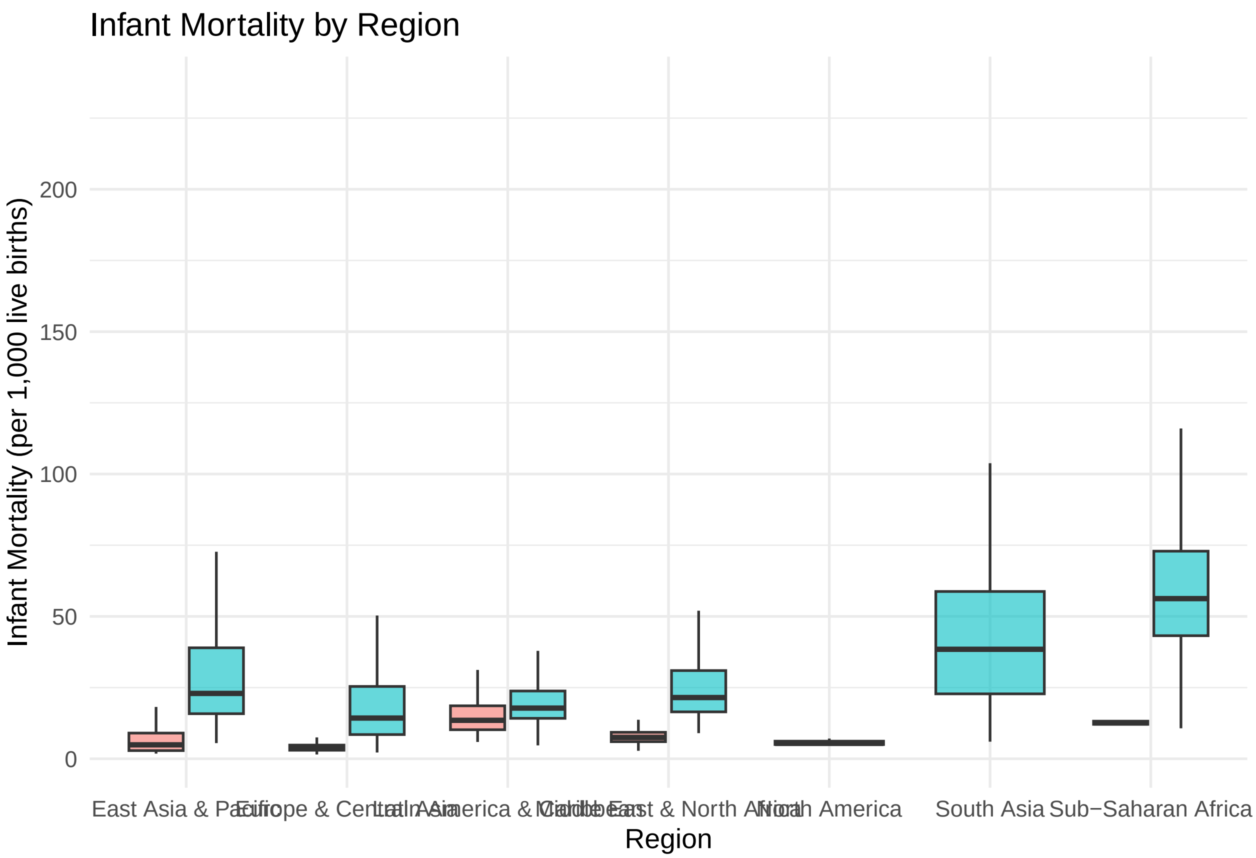
```
##
## Call:
## lm(formula = InfantMortality ~ region, data = data_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -46.771  -7.603  -3.485   6.497 177.429
##
## Coefficients:
```

```
##                                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)                        22.8031     0.6723  33.918  < 2e-16 ***
## regionEurope & Central Asia       -13.8817     0.8428 -16.470  < 2e-16 ***
## regionLatin America & Caribbean    -3.6180     0.9271  -3.902 9.69e-05 ***
## regionMiddle East & North Africa   -3.4102     1.0393  -3.281  0.00104 **
## regionNorth America               -17.2031     2.6407  -6.515 8.22e-11 ***
## regionSouth Asia                   18.6433     1.4498  12.859  < 2e-16 ***
## regionSub-Saharan Africa           34.6682     0.8609  40.268  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 16.55 on 3922 degrees of freedom
## Multiple R-squared:  0.5581, Adjusted R-squared:  0.5574
## F-statistic: 825.4 on 6 and 3922 DF,  p-value: < 2.2e-16
```

```
car::Anova(region_model, type = 3)
```

```
## Anova Table (Type III tests)
##
## Response: InfantMortality
##              Sum Sq   Df F value    Pr(>F)
## (Intercept)  315110    1 1150.44 < 2.2e-16 ***
## region      1356561    6  825.45 < 2.2e-16 ***
## Residuals   1074251 3922
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# plot
ggplot(data_clean, aes(x = region, y = InfantMortality, fill = income_binary)) +
  geom_boxplot(alpha = 0.6, outlier.shape = NA) +
  labs(
    x = "Region",
    y = "Infant Mortality (per 1,000 live births)",
    title = "Infant Mortality by Region"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



Describe the Result: A linear regression was used to assess whether Infant Mortality rates significantly differ by regions. The overall ANOVA revealed a significant effect of region on Infant Mortality, indicating that Infant Mortality rates differ substantially across global regions. Follow-up comparisons from the model summary show that regions such as Sub-Saharan Africa and South Asia have significantly higher Infant Mortality rates compared to the reference region (Europe & Central Asia). –

Question 4

You propose a more complex idea to your director: "Director, maybe the effect of being 'High Income' is different in 'Sub-Saharan Africa' than it is in 'Europe & Central Asia'. Maybe income status matters more in some regions?"

- Build a factorial model using lm() that includes YOUR_DV as the dependent variable and both income_binary and region (along with their interaction) as independent variables.
- Use car::Anova(your_model, type = 3) to get the F-tests.
- Visualize: Create an interaction plot (stat_summary(fun = mean, geom = "line")) showing the relationship between income_binary and YOUR_DV, with different colored lines for region. Add something to this plot to impress your director (e.g., error bars, jittered data, etc.)
- Look at your car::Anova() output. Is the income_binary:region interaction significant?
- Look at your interaction plot. Are the lines parallel? In plain English, describe the ANOVA output and the the plot to your director.

```
# Model 4: The "Interaction" Model (The Full Factorial Model)
# Assumes the effect of 'dose' depends on 'supp'.
# model_4 <- lm(len ~ supp * dose, data = ToothGrowth)
```

```r
table(data_clean$region, data_clean$income_binary)
```

```
##
##                            High Income Not High Income
##   East Asia & Pacific              168             438
##   Europe & Central Asia            714             346
##   Latin America & Caribbean        189             483
##   Middle East & North Africa       168             268
##   North America                     42               0
##   South Asia                         0             166
##   Sub-Saharan Africa                21             926
```

```r
# we have some regions like north america and south asia that have 0 low/high
# income countries so we need to filter
data_interact <- data_clean %>%
  filter(!(region %in% c("North America", "South Asia"))) %>%
  droplevels()

# need to do predictor * predictor for factorial model
region_model2 <- lm(InfantMortality ~ region * income_binary,
                    data = data_interact)
summary(region_model2)
```

```
##
## Call:
## lm(formula = InfantMortality ~ region * income_binary, data = data_interact)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -47.788  -7.489  -1.282   4.318 176.412
##
## Coefficients:
##                                                        Estimate
## (Intercept)                                              8.5393
## regionEurope & Central Asia                             -4.1575
## regionLatin America & Caribbean                          7.1581
## regionMiddle East & North Africa                        -0.7762
## regionSub-Saharan Africa                                 4.1131
## income_binaryNot High Income                            19.7349
## regionEurope & Central Asia:income_binaryNot High Income      -5.8274
## regionLatin America & Caribbean:income_binaryNot High Income  -14.8824
## regionMiddle East & North Africa:income_binaryNot High Income  -0.8148
## regionSub-Saharan Africa:income_binaryNot High Income         26.1005
##                                                        Std. Error
## (Intercept)                                                1.1319
## regionEurope & Central Asia                                1.2581
## regionLatin America & Caribbean                            1.5557
## regionMiddle East & North Africa                           1.6008
## regionSub-Saharan Africa                                   3.3958
## income_binaryNot High Income                               1.3314
## regionEurope & Central Asia:income_binaryNot High Income       1.6420
## regionLatin America & Caribbean:income_binaryNot High Income   1.8323
## regionMiddle East & North Africa:income_binaryNot High Income  1.9640
## regionSub-Saharan Africa:income_binaryNot High Income          3.5008
```

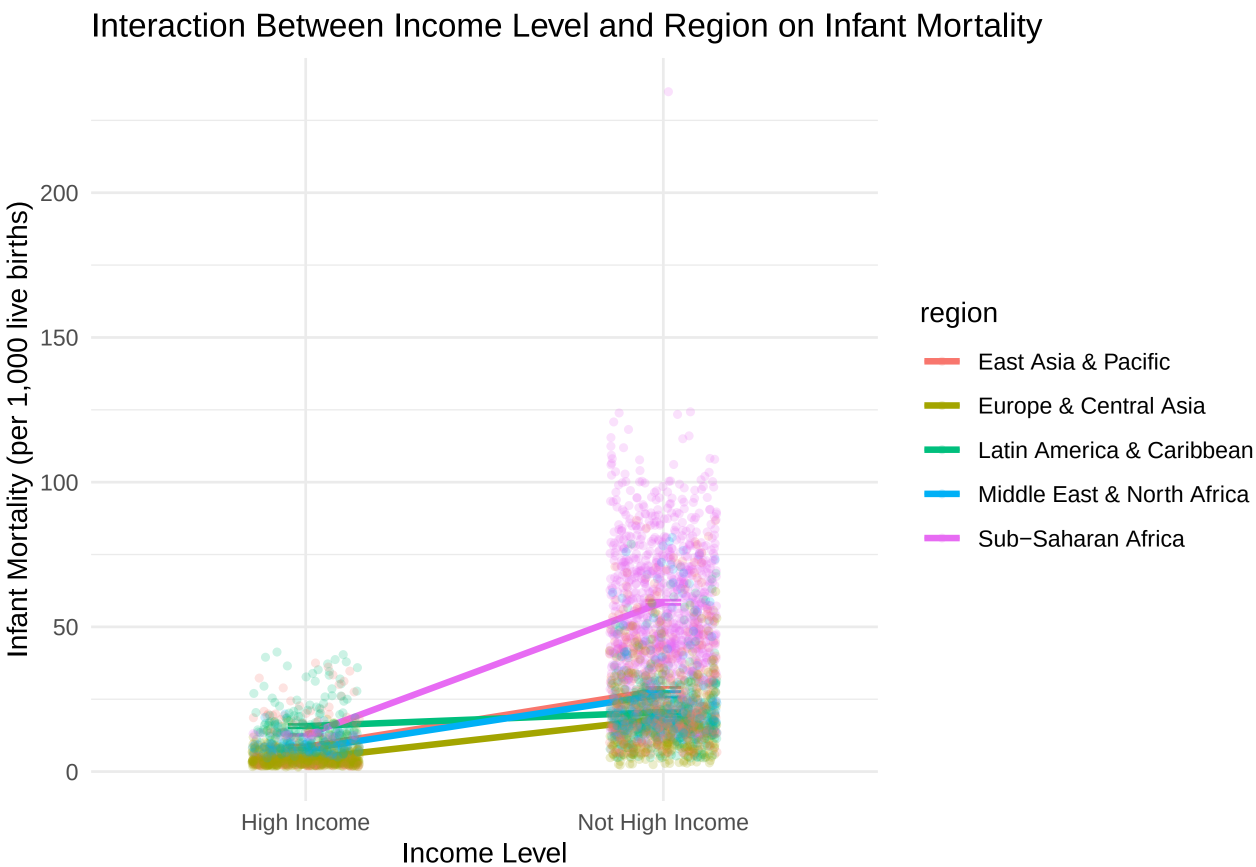
```
##                                                                 t value Pr(>|t|)
## (Intercept)                                                      7.544 5.70e-14
## regionEurope & Central Asia                                     -3.305 0.000960
## regionLatin America & Caribbean                                  4.601 4.34e-06
## regionMiddle East & North Africa                                -0.485 0.627790
## regionSub-Saharan Africa                                         1.211 0.225882
## income_binaryNot High Income                                    14.822  < 2e-16
## regionEurope & Central Asia:income_binaryNot High Income        -3.549 0.000392
## regionLatin America & Caribbean:income_binaryNot High Income    -8.122 6.17e-16
## regionMiddle East & North Africa:income_binaryNot High Income  -0.415 0.678256
## regionSub-Saharan Africa:income_binaryNot High Income            7.456 1.11e-13
##
## (Intercept)                                                     ***
## regionEurope & Central Asia                                     ***
## regionLatin America & Caribbean                                 ***
## regionMiddle East & North Africa
## regionSub-Saharan Africa
## income_binaryNot High Income                                    ***
## regionEurope & Central Asia:income_binaryNot High Income        ***
## regionLatin America & Caribbean:income_binaryNot High Income    ***
## regionMiddle East & North Africa:income_binaryNot High Income
## regionSub-Saharan Africa:income_binaryNot High Income           ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.67 on 3711 degrees of freedom
## Multiple R-squared:  0.6491, Adjusted R-squared:  0.6483
## F-statistic: 762.8 on 9 and 3711 DF,  p-value: < 2.2e-16
```

```r
car::Anova(region_model2, type = 3)
```

```
## Anova Table (Type III tests)
##
## Response: InfantMortality
##                       Sum Sq   Df F value    Pr(>F)
## (Intercept)            12250    1  56.912 5.699e-14 ***
## region                 20296    4  23.572 < 2.2e-16 ***
## income_binary          47291    1 219.702 < 2.2e-16 ***
## region:income_binary   38099    4  44.249 < 2.2e-16 ***
## Residuals             798800 3711
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# plot w jitter and interaction effect
ggplot(data_interact, aes(x = income_binary, y = InfantMortality, color = region,
                         group = region)) +
  stat_summary(fun = mean, geom = "line", size = 1.2) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.1) +
  geom_jitter(alpha = 0.2, width = 0.15, height = 0, size = 1) +
  labs(
    title = "Interaction Between Income Level and Region on Infant Mortality",
    x = "Income Level",
    y = "Infant Mortality (per 1,000 live births)"
  ) +
  theme_minimal() +
```

```r
  theme(legend.position = "right")
```



Describe the Result: A factorial ANOVA was used to assess whether infant mortality rates significantly differ by regions and income status. It's important to note that the countries of North America and South Asia were excluded from the analyses, because of aliased coefficients. There were significant main effects of both region and income status ($p < .001$), suggesting that infant mortality rates differ across both geographic regions and income groups. The income × region interaction was also significant ($p < .001$), indicating that the effect of income status on Infant Mortality does significantly differ by region. The lines of the interaction plot are not parallel, suggesting the impact of income on Infant Mortality varies across regions, and income appears to have a stronger effect on infant mortality in Sub-Saharan Africa and Asia than in Europe. –

Question 5

As with all data analyses, this is getting complicated. Your director asks you to build one model for a single year to understand the key relationships. For 2019, what really predicts [Your DV]? Is it [Your IV1], [Your IV2], [Your IV3], or region?"

- Filter: Create a new dataframe data_2019 that only contains data where year == 2019.
- Inspect: Inspect the distribution and interrelationships among your variables using graphs.
- Model: Build a "main effects" only multiple regression model using lm().
- Use car::Anova(your_model, type = 3) to get the results.
- Look at the car::Anova() output. When controlling for all factors simultaneously (Type 3 MSE), which ones are significant?
- Synthesis: Look at the summary() output. Compare the coefficient for region here to the one you got in Question 4. Has its significance or size changed? Why might that be? (Hint: What other variable in

your model is region likely correlated with?)

- Repeat this analysis for year == 2001 and compare the two sets of results.

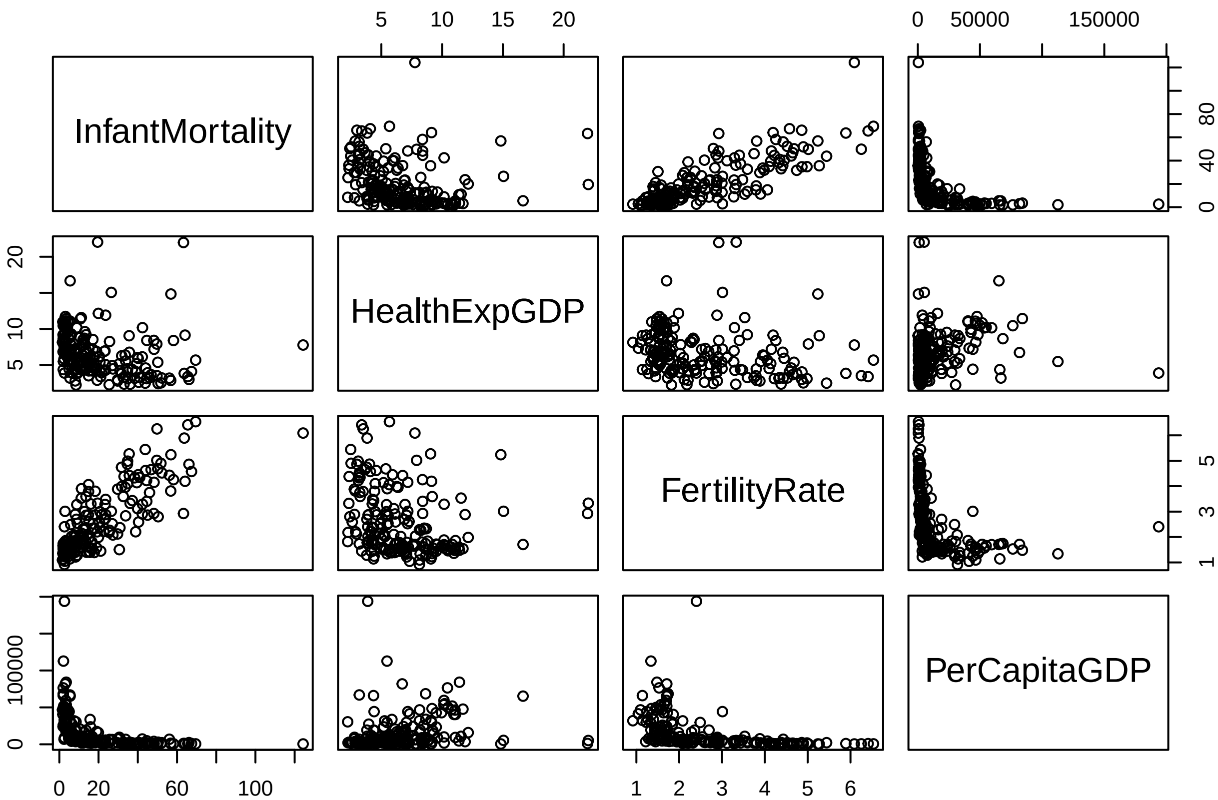
```
data_2019 <- data_clean %>%
  filter(year == 2019)

summary(data_2019)
```

```
##    country              iso2c              iso3c                year     
##  Length:187         Length:187         Length:187         Min.   :2019  
##  Class :character   Class :character   Class :character   1st Qu.:2019  
##  Mode  :character   Mode  :character   Mode  :character   Median :2019  
##                                                           Mean   :2019  
##                                                           3rd Qu.:2019  
##                                                           Max.   :2019  
##     status           lastupdated        InfantMortality   HealthExpGDP  
##  Length:187         Length:187         Min.   :  1.60   Min.   : 2.228  
##  Class :character   Class :character   1st Qu.:  5.50   1st Qu.: 4.269  
##  Mode  :character   Mode  :character   Median : 13.70   Median : 6.027  
##                                        Mean   : 20.49   Mean   : 6.579  
##                                        3rd Qu.: 32.20   3rd Qu.: 8.391  
##                                        Max.   :124.30   Max.   :22.030  
##  FertilityRate    PerCapitaGDP        region            capital         
##  Min.   :0.918   Min.   :   210.2   Length:187         Length:187        
##  1st Qu.:1.589   1st Qu.:  2188.3   Class :character   Class :character  
##  Median :2.142   Median :  6205.1   Mode  :character   Mode  :character  
##  Mean   :2.596   Mean   : 15540.1                                        
##  3rd Qu.:3.326   3rd Qu.: 19018.9                                        
##  Max.   :6.537   Max.   :193746.8                                        
##   longitude           latitude            income            lending         
##  Length:187         Length:187         Length:187         Length:187        
##  Class :character   Class :character   Class :character   Class :character  
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character  
##                                                                             
##                                                                             
##                                                                             
##          income_binary
##  High Income    : 62  
##  Not High Income:125  
##                       
##                       
##                       
##                       
```

```
# correlations
data_2019 %>%
  select(InfantMortality, HealthExpGDP, FertilityRate, PerCapitaGDP) %>%
  cor(use = "pairwise.complete.obs") %>%
  round(2)
```

```
##                 InfantMortality HealthExpGDP FertilityRate PerCapitaGDP
## InfantMortality            1.00        -0.21          0.82        -0.47
## HealthExpGDP              -0.21         1.00         -0.24         0.22
## FertilityRate              0.82        -0.24          1.00        -0.42
## PerCapitaGDP              -0.47         0.22         -0.42         1.00
```

```r
# visuals
pairs(~ InfantMortality + HealthExpGDP + FertilityRate + PerCapitaGDP, data = data_2019)
```




```r
model_2019 <- lm(InfantMortality ~ HealthExpGDP + FertilityRate + PerCapitaGDP + region,
                 data = data_2019)
summary(model_2019)
```

```
##
## Call:
## lm(formula = InfantMortality ~ HealthExpGDP + FertilityRate +
##     PerCapitaGDP + region, data = data_2019)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -29.968  -4.613  -0.681   3.530  62.860
##
## Coefficients:
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                    -4.706e+00  3.661e+00  -1.285 0.200385
## HealthExpGDP                    1.994e-01  2.538e-01   0.785 0.433253
## FertilityRate                   8.946e+00  9.625e-01   9.294  < 2e-16 ***
## PerCapitaGDP                   -8.554e-05  3.754e-05  -2.279 0.023875 *
## regionEurope & Central Asia    -3.551e+00  2.516e+00  -1.412 0.159798
## regionLatin America & Caribbean  2.564e+00  2.653e+00   0.966 0.335150
## regionMiddle East & North Africa -2.940e+00  2.968e+00  -0.991 0.323195
## regionNorth America            -2.491e+00  7.747e+00  -0.322 0.748181
## regionSouth Asia                9.697e+00  4.076e+00   2.379 0.018426 *
## regionSub-Saharan Africa        1.016e+01  2.950e+00   3.443 0.000717 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.12 on 177 degrees of freedom
## Multiple R-squared:  0.7364, Adjusted R-squared:  0.723
## F-statistic: 54.94 on 9 and 177 DF,  p-value: < 2.2e-16
```

```
car::Anova(model_2019, type = 3)
```

```
## Anova Table (Type III tests)
##
## Response: InfantMortality
##                Sum Sq  Df F value    Pr(>F)
## (Intercept)     169.1   1  1.6519 0.2003849
## HealthExpGDP     63.2   1  0.6169 0.4332528
## FertilityRate  8845.2   1 86.3850 < 2.2e-16 ***
## PerCapitaGDP    531.7   1  5.1928 0.0238745 *
## region         2910.5   6  4.7375 0.0001681 ***
## Residuals     18123.5 177
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
data_2001 <- data_clean %>%
  filter(year == 2001)
summary(data_2001)
```

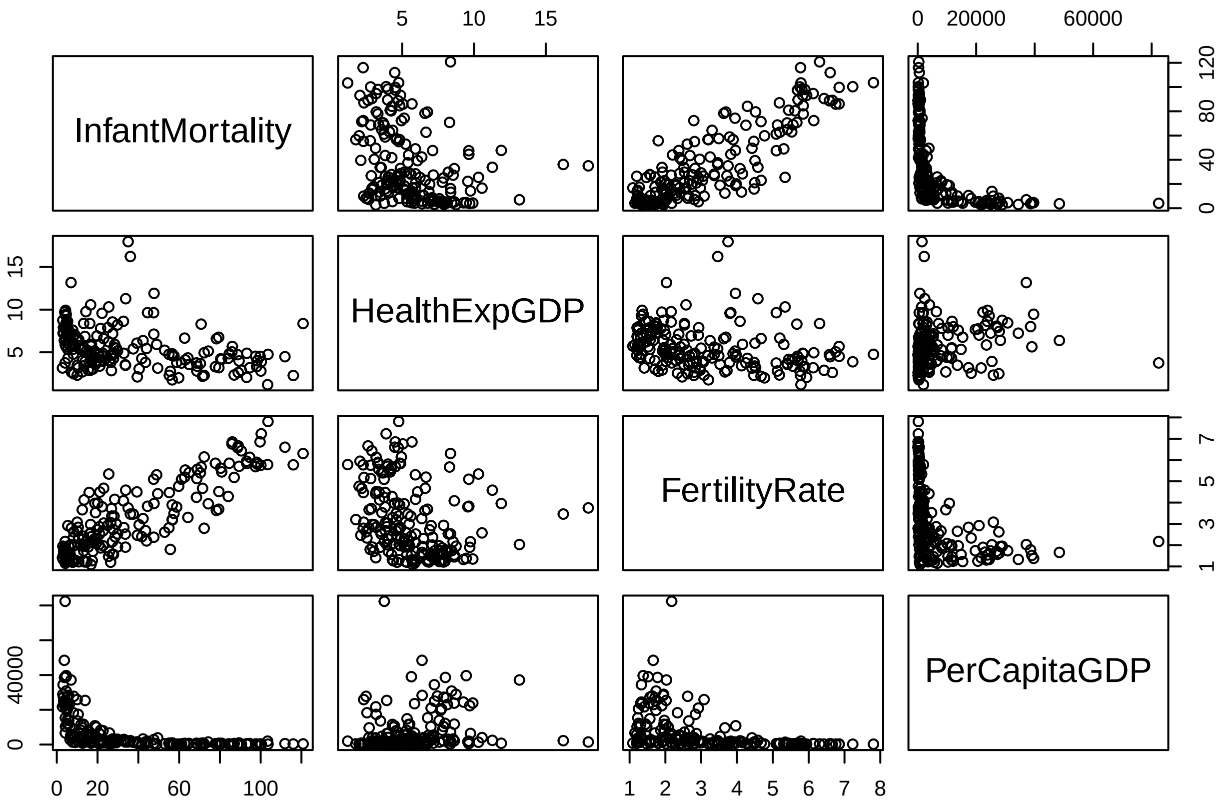
```
##    country             iso2c              iso3c                year
##  Length:184         Length:184         Length:184         Min.   :2001
##  Class :character   Class :character   Class :character   1st Qu.:2001
##  Mode  :character   Mode  :character   Mode  :character   Median :2001
##                                                           Mean   :2001
##                                                           3rd Qu.:2001
##                                                           Max.   :2001
##     status          lastupdated        InfantMortality  HealthExpGDP
##  Length:184         Length:184         Min.   :  2.8   Min.   : 1.203
##  Class :character   Class :character   1st Qu.:  9.8   1st Qu.: 3.818
##  Mode  :character   Mode  :character   Median : 24.0   Median : 5.087
##                                        Mean   : 35.7   Mean   : 5.578
##                                        3rd Qu.: 56.9   3rd Qu.: 7.004
##                                        Max.   :120.8   Max.   :17.965
##  FertilityRate   PerCapitaGDP       region             capital
##  Min.   :1.092   Min.   :  118.5   Length:184         Length:184
##  1st Qu.:1.740   1st Qu.:  648.2   Class :character   Class :character
##  Median :2.679   Median : 1853.1   Mode  :character   Mode  :character
##  Mean   :3.175   Mean   : 7001.7
##  3rd Qu.:4.430   3rd Qu.: 7344.3
##  Max.   :7.810   Max.   :82402.8
##   longitude           latitude            income             lending
##  Length:184         Length:184         Length:184         Length:184
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##          income_binary
##  High Income    : 62
```

```
##   Not High Income:122
##
##
##
##
```

```r
# correlations
data_2001 %>%
  select(InfantMortality, HealthExpGDP, FertilityRate, PerCapitaGDP) %>%
  cor(use = "pairwise.complete.obs") %>%
  round(2)
```

```
##                   InfantMortality HealthExpGDP FertilityRate PerCapitaGDP
## InfantMortality              1.00        -0.34          0.86        -0.51
## HealthExpGDP                -0.34         1.00         -0.27         0.23
## FertilityRate                0.86        -0.27          1.00        -0.43
## PerCapitaGDP                -0.51         0.23         -0.43         1.00
```

```r
# visuals
pairs(~ InfantMortality + HealthExpGDP + FertilityRate + PerCapitaGDP, data = data_2001)
```




```r
model_2001 <- lm(InfantMortality ~ HealthExpGDP + FertilityRate + PerCapitaGDP + region,
                 data = data_2001)
summary(model_2001)
```

```
##
## Call:
## lm(formula = InfantMortality ~ HealthExpGDP + FertilityRate +
##     PerCapitaGDP + region, data = data_2001)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -36.081  -7.293  -0.980   7.074  40.493
##
## Coefficients:
##                                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)                    0.2542696  4.8761549   0.052 0.958473
## HealthExpGDP                  -1.2105394  0.4242604  -2.853 0.004852 **
## FertilityRate                 12.7474773  1.0406689  12.249  < 2e-16 ***
## PerCapitaGDP                  -0.0004966  0.0001024  -4.849 2.73e-06 ***
## regionEurope & Central Asia    7.8382182  3.5238519   2.224 0.027412 *
## regionLatin America & Caribbean -1.7963624  3.4884770  -0.515 0.607247
## regionMiddle East & North Africa -6.1684240  3.9567038  -1.559 0.120818
## regionNorth America           11.4650870 10.3260642   1.110 0.268399
## regionSouth Asia              14.7393232  5.7311735   2.572 0.010953 *
## regionSub-Saharan Africa      13.7874105  4.0501869   3.404 0.000824 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.39 on 174 degrees of freedom
## Multiple R-squared:  0.8262, Adjusted R-squared:  0.8173
## F-statistic: 91.93 on 9 and 174 DF,  p-value: < 2.2e-16
```

```r
car::Anova(model_2001, type = 3)
```

```
## Anova Table (Type III tests)
##
## Response: InfantMortality
##                Sum Sq  Df  F value    Pr(>F)
## (Intercept)       0.5   1   0.0027  0.958473
## HealthExpGDP   1459.8   1   8.1413  0.004852 **
## FertilityRate 26904.4   1 150.0456 < 2.2e-16 ***
## PerCapitaGDP   4216.8   1  23.5171 2.733e-06 ***
## region         8156.6   6   7.5815 3.079e-07 ***
## Residuals     31199.6 174
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Describe the Result: Fertility rate and region are significant. The coefficients for sub-saharan africa and asia are ($b = 9.697e+00$, $b = 1.016e+01$) respectively. The significance for region has stayed the same, but the size has decreased ($p = .024$). This is most likely due to the relationship between region and per capita GDP. In 2001, the results have replicated, showing that from 2001 to 2019, measures of infant mortality by health expenditure GDP, fertility rate, per capita GDP, and region have stayed relatively stable. –

Question 6

You realize a major flaw in all your previous models (other than Question 5). Your data_clean dataframe has multiple entries for each country (from 2000-2020). These observations are not independent! You must use a mixed-effects model.

- Model 1 (Fixed effects only): Using the full data_clean dataset, build an lm() model. Predict YOUR_DV with the fixed effects of YOUR_IV1, YOUR_IV2, YOUR_IV3, and year.
- Model 2 (Random Intercept): Using the full data_clean dataset, build an lmer() model. Predict YOUR_DV with the fixed effects of YOUR_IV1, YOUR_IV2, YOUR_IV3, and year. Include a random intercept for country.
- Model 3 (Add Random Slope): Using the full data_clean dataset, build an lmer() model. Predict

YOUR__DV with the fixed effects of YOUR__IV1, YOUR__IV2, YOUR__IV3, and year. Add a random slope for year by country. (This model will have both random intercepts and slopes)

- Compare Models: Use anova(model_1, model_2) to see which is a better fit.
- Visualize: Check the normality of residuals and random effects of your best model.
- Based on your anova() comparison, which model is better? How do you know?
- Look at the summary() of your best and final model.
- Fixed Effects: What is the coefficient for year? What does this tell you about the general trend of [Your DV] worldwide, even after accounting for your other IVs?
- Random Effects: Look at the "Random effects" variance table.
  - What does the (Intercept) variance for country tell you?
  - What does the year variance for country (the random slope) tell you?
- Final Synthesis: Write a final, 3-4 sentence paragraph to your director. What are the key drivers of [Your DV]? And what important, real-world story (about the differences between countries) did our simple models completely miss that our LMM was able to find?

```r
# fixed effects only
model_1 <- lm(InfantMortality ~ HealthExpGDP + FertilityRate + PerCapitaGDP
              + year, data = data_clean)

# random intercept (country)
model_2 <- lmer(InfantMortality ~ HealthExpGDP + FertilityRate + PerCapitaGDP
                + year + (1 | country), data = data_clean, REML = FALSE)

# random intercept + random slope for year by country
model_3 <- lmer(InfantMortality ~ HealthExpGDP + FertilityRate + PerCapitaGDP +
                year + (1 + year | country), data = data_clean, REML = FALSE)

# compare model_1 vs model_2
# anova(model_1, model_2) does not work bc we can't compare lm to lmer so
# let's use AIC to compare them instead (lower = better)
AIC(model_1)
```

```
## [1] 30962.59
```

```r
AIC(model_2)
```

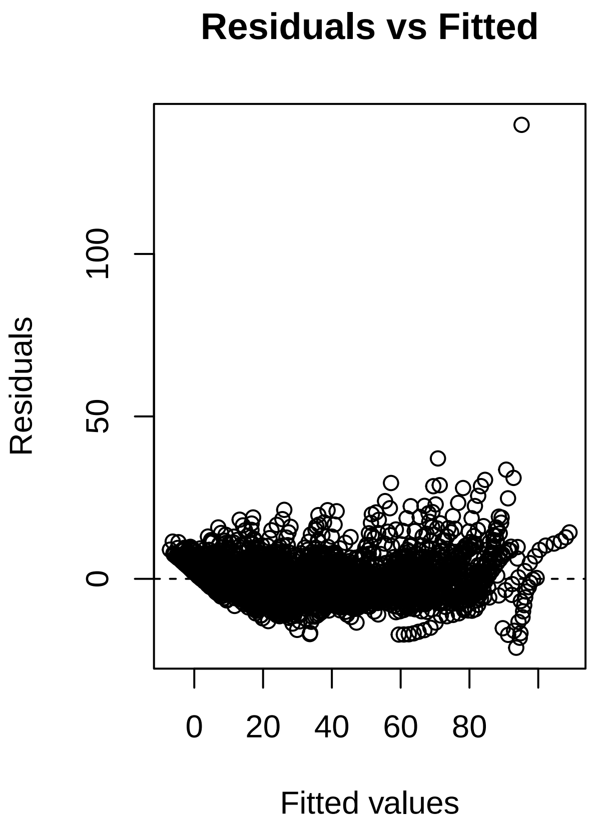
```
## [1] 25906.13
```

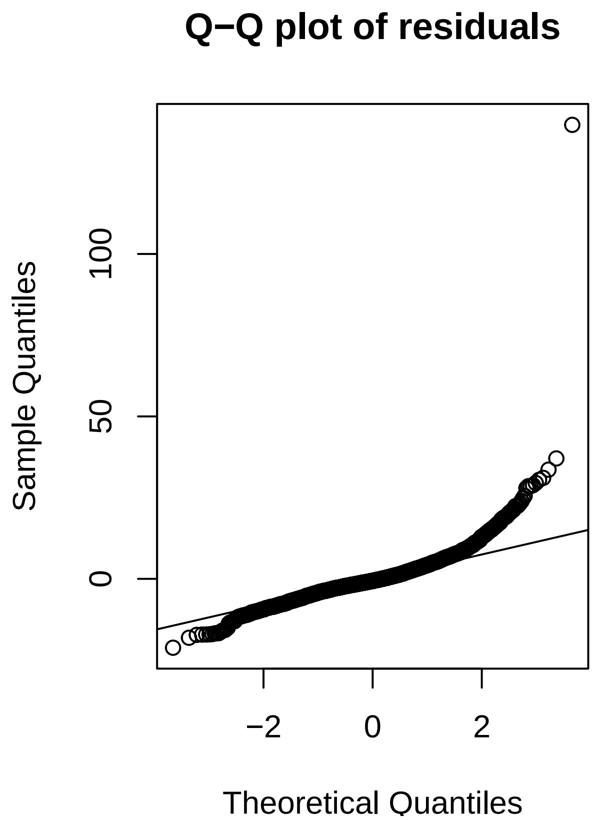
```r
anova(model_2, model_3)
```

```
## Data: data_clean
## Models:
## model_2: InfantMortality ~ HealthExpGDP + FertilityRate + PerCapitaGDP + year + (1 | country)
## model_3: InfantMortality ~ HealthExpGDP + FertilityRate + PerCapitaGDP + year + (1 + year | country)
##         npar   AIC   BIC logLik -2*log(L) Chisq Df Pr(>Chisq)
## model_2    7 25906 25950 -12946     25892
## model_3    9 25945 26002 -12964     25927     0  2          1
```

```r
# it seems like model 2 is best!
best_model <- model_2

# residuals vs fitted
par(mfrow = c(1,2))
```

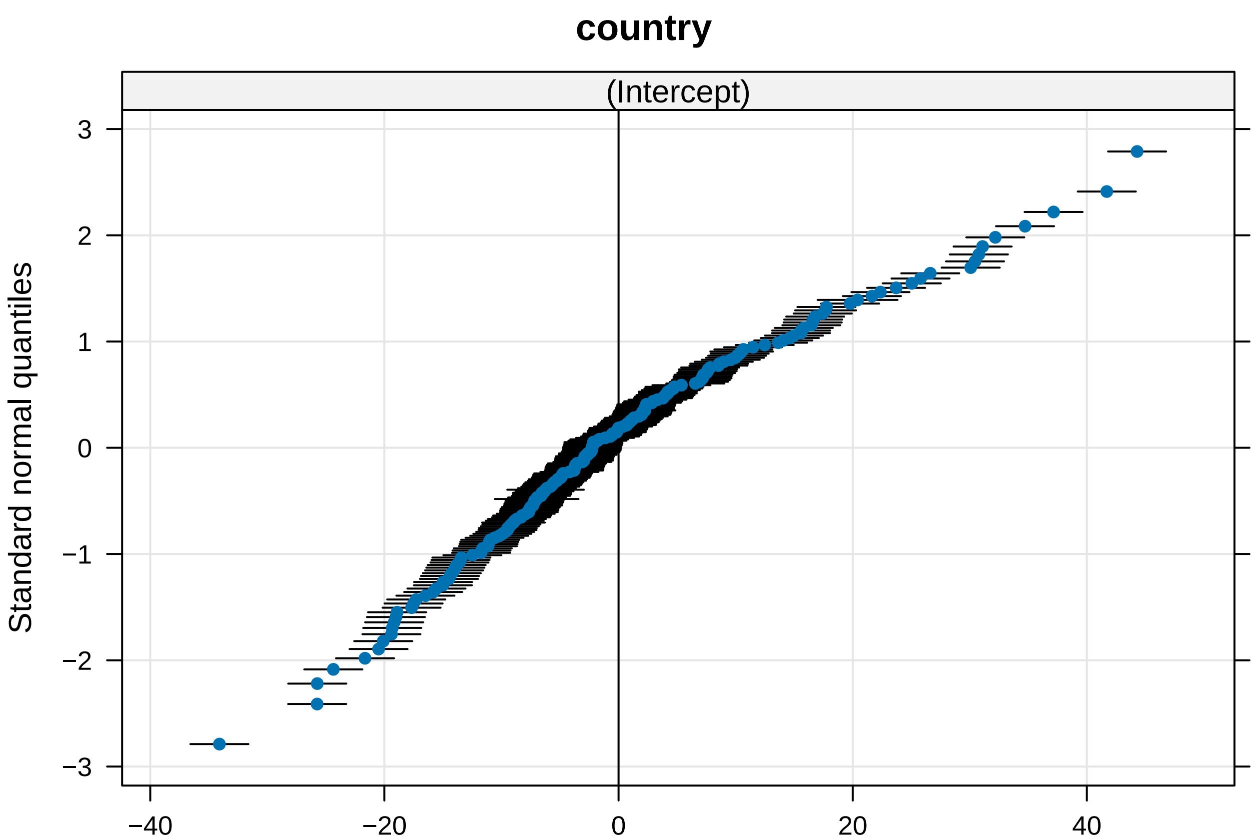
```r
plot(fitted(best_model), resid(best_model),
     xlab = "Fitted values", ylab = "Residuals",
     main = "Residuals vs Fitted")
abline(h = 0, lty = 2)

# normal Q-Q of residuals
qqnorm(resid(best_model), main = "Q-Q plot of residuals")
qqline(resid(best_model))
```






```r
# random effects check
re <- ranef(best_model, condVar = TRUE)
# Create Q-Q plots for each random effect
# condVar = TRUE adds the confidence bands
qqmath(ranef(best_model, condVar = TRUE))
```

```
## $country
```




```r
summary(best_model)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: InfantMortality ~ HealthExpGDP + FertilityRate + PerCapitaGDP +
##     year + (1 | country)
##    Data: data_clean
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   25906.1   25950.1  -12946.1   25892.1      3922
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.6396 -0.4899 -0.1090  0.4066 23.9884
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  country  (Intercept) 184.34   13.577
##  Residual              33.94    5.825
## Number of obs: 3929, groups:  country, 189
##
## Fixed effects:
##                 Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)    1.175e+03  4.229e+01  3.336e+03  27.779  < 2e-16 ***
## HealthExpGDP  -7.021e-02  8.321e-02  3.929e+03  -0.844    0.399
## FertilityRate  1.053e+01  3.218e-01  1.881e+03  32.736  < 2e-16 ***
## PerCapitaGDP   1.208e-04  1.657e-05  3.314e+03   7.288  3.9e-13 ***
## year          -5.868e-01  2.088e-02  3.389e+03 -28.101  < 2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) HlEGDP FrtltR PrCGDP
## HelthExpGDP  0.238
## FertilityRt -0.547 -0.010
## PerCapitGDP  0.399  0.011 -0.128
## year        -0.999 -0.251  0.529 -0.404
## fit warnings:
## Some predictor variables are on very different scales: consider rescaling
```

Describe the Result: We know that model 2 is the best because it has the lowest AIC. The coefficient for year is (b = -.59). This tells us that controlling for health expenditure, fertility rate, and GDP per capita, Infant Mortality has been decreasing by approximately -.59 year units per year worldwide. The variance for country is 184.34 and tells us how much countries differ in their baseline infant mortality after accounting for the fixed predictors. Because the intercept variance is large, we know that there is higher heterogeneity between countries baseline infant mortality rates. The year variance for country 33.94, indicating that countries don't differ strongly in whether they're improving faster or slower over time.

Final Synthesis: Write a final, 3-4 sentence paragraph to your director. What are the key drivers of [Your DV]? And what important, real-world story (about the differences between countries) did our simple models completely miss that our LMM was able to find? After accounting for fertility rate, health expenditure GDP, and GDP per capita, the linear mixed model shows that average infant mortality declined by about -.59 (b(year) = -.59) deaths per 1,000 live births per year worldwide, holding other predictors constant. It seems that higher fertility is associated with higher infant mortality, while greater health expenditure and higher GDP per capita predict lower infant mortality, making these the key drivers of infant mortality. Crucially, the random effects tell a different story as countries differ substantially in baseline infant mortality. There is a large country intercept variance (var = 184.34) and, when we include random slopes, in their rates of improvement over time (var(year) = 33.94) too. While global averages show improvement, some countries started much worse and others improved much faster or slower. The LMM reveals that national-level heterogeneity in both baseline healthcare differences and rate of improvement is an important part of the real-world story that our simple pooled models missed. –

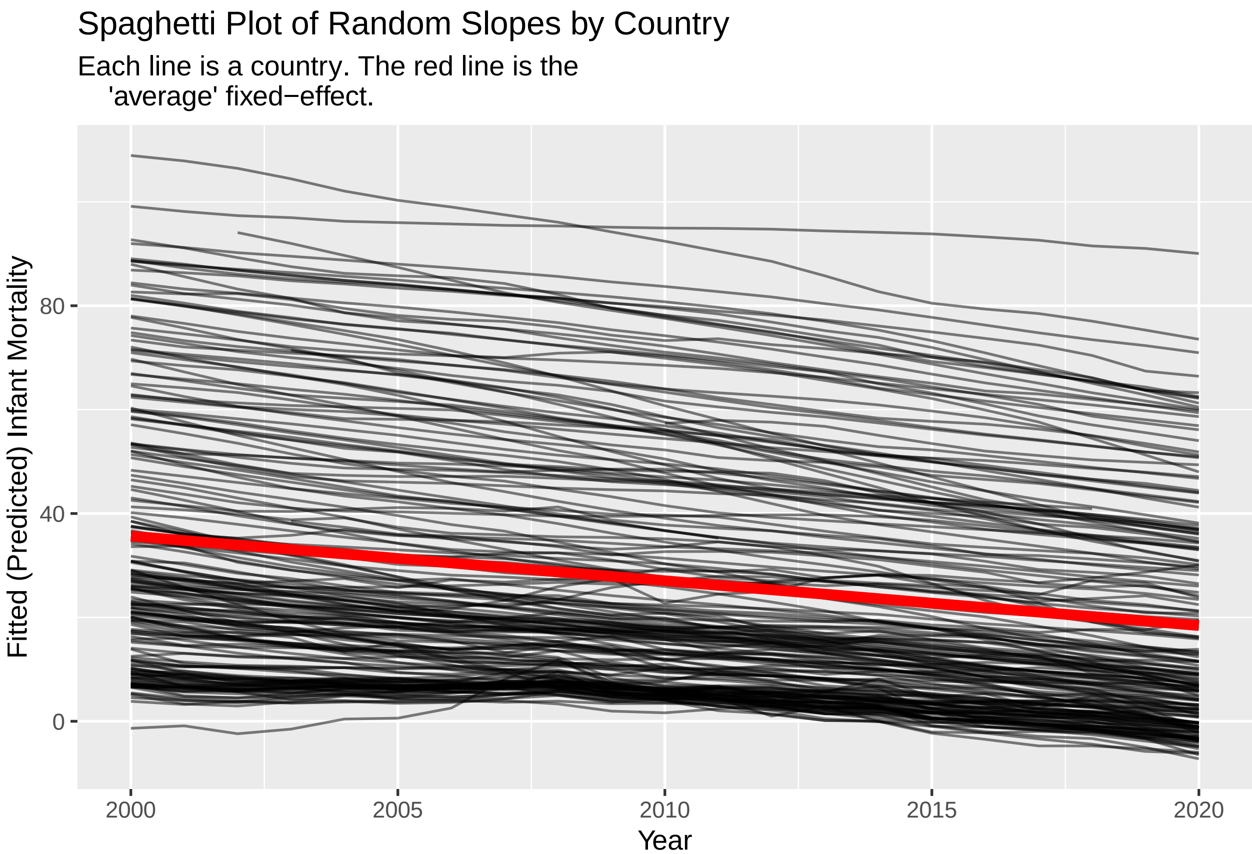
Question 7

Intrigued by your LMM analysis, the director asks you to create a spaghetti plot of your random slopes model to help make the abstract concept of "random slope variance" from lmer() more concrete. This "spaghetti plot" plots the fitted model line for every single country. It's the best way to visualize what the LMM is doing.

- First, you fit the full lmer() model with a random slope for year (you just did this in Question 6).
- Then, you use the augment() (from broom.mixed) to get the .fitted (predicted) value for each country at each time point. Plotting these fitted lines shows the unique trajectory you've modeled for each country.

```r
# random intercept + random slope for year by country
model_3 <- lmer(InfantMortality ~ HealthExpGDP + FertilityRate + PerCapitaGDP +
                year + (1 + year | country), data = data_clean, REML = FALSE)

fitted_data <- augment(model_3)

ggplot(fitted_data, aes(x = year, y = .fitted, group = country)) +
  geom_line(alpha = 0.5) +
  geom_smooth(aes(group = 1), method = "lm", se = FALSE, color = "red",
              linewidth = 2) +
```

```
  labs(
    title = "Spaghetti Plot of Random Slopes by Country",
    subtitle = "Each line is a country. The red line is the
    'average' fixed-effect.",
    x = "Year",
    y = "Fitted (Predicted) Infant Mortality"
  )
```




```
# Use broom.mixed::augment() to get fitted values
#fitted_data <- augment(lmm_model)

#Use GGplot to plot the fitted slopes
#ggplot(fitted_data, aes(x = year, y = .fitted, group = country)) +
  # The geom_line() connects the dots for each "group" (country)
#  geom_line(alpha = 0.5) +
  # Add the "average" (fixed effect) line in red
#  geom_smooth(aes(group = 1), method = "lm", se = FALSE, color = "red", linewidth = 2) +
#  labs(
#    title = "The 'Spaghetti Plot' of Random Slopes",
#    subtitle = "Each line is a country. The red line is the 'average' fixed effect.",
#    x = "Year",
#    y = "Fitted (Predicted) DV"
#  )
```

Summarize this plot in 1-2 sentences: The spaghetti plot shows that while most countries follow a general downward trend in infant mortality over time, the slopes vary across countries. This is because each country has its own unique rate of change (random slopes) in infant mortality over the years, around the average trend shown by the red fixed-effect line. –

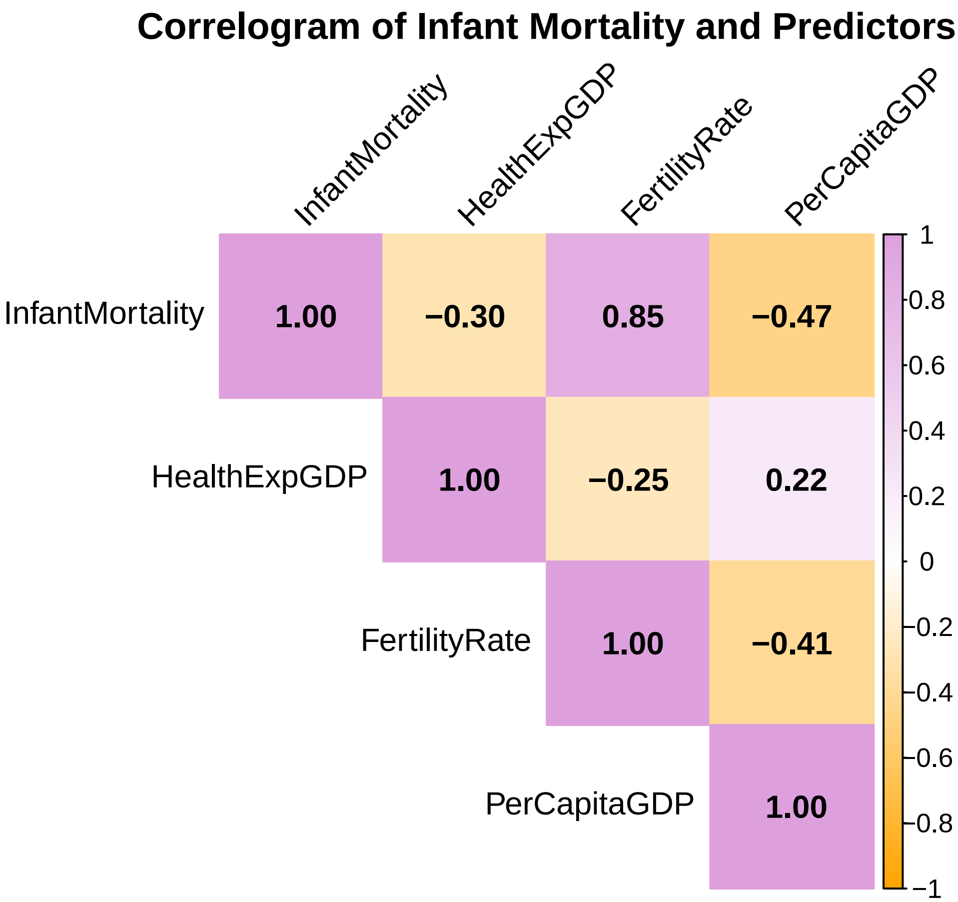
Question 8

The director has asked you to suggest an analysis that might help to understand the interrelationships between your variables. Conduct and present the results of that analysis in the space below.

```r
library(corrplot)
# let's make a correlogram for our key vars
interrel <- data_clean %>%
  select(InfantMortality, HealthExpGDP, FertilityRate, PerCapitaGDP)

# correlation matrix
cor_matrix <- cor(interrel, use = "pairwise.complete.obs")

cor_matrix
```

```
##                 InfantMortality HealthExpGDP FertilityRate PerCapitaGDP
## InfantMortality       1.0000000   -0.2965161     0.8487852   -0.4665611
## HealthExpGDP         -0.2965161    1.0000000    -0.2518927    0.2247340
## FertilityRate         0.8487852   -0.2518927     1.0000000   -0.4059618
## PerCapitaGDP         -0.4665611    0.2247340    -0.4059618    1.0000000
```

```r
corrplot(cor_matrix,
         method = "color",          # colored squares
         type = "upper",            # upper triangle
         addCoef.col = "black",     # add correlation coefficients
         tl.col = "black",          # label color
         tl.srt = 45,               # rotate labels
         col = colorRampPalette(c("orange", "white", "plum"))(200),
         title = "Correlogram of Infant Mortality and Predictors",
         mar = c(0,0,2,0))
```



Describe the Result: The correlogram shows that infant mortality is moderately, negatively correlated with both health expenditure GDP and GDP per capita, but strongly, positively correlated with fertility rate. These relationships suggest that countries with higher income and healthcare investment tend to have lower infant mortality, while higher fertility rates can also produce higher infant mortality. –